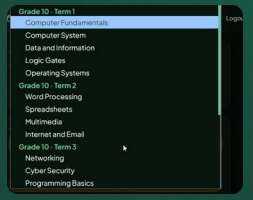
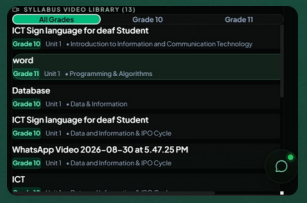
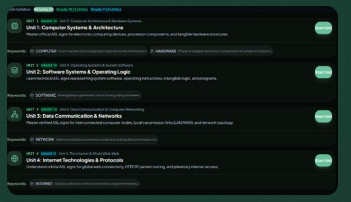
Accessible Emotion-Aware E-Learning Platform for Deaf Students Studying O/L Information Technology

Research Group: R26-IT-018



Accessible Emotion-Aware E-Learning Platform for Deaf Students Studying O/L Information Technology

Panel Comment	How We Addressed It	Evidence
Make chatbot responses and recommendations syllabus-aware and relevant to learning needs.	Added Grade 10/11 ICT topic selection to guide responses, learning plans, and mode suggestions.	
Align video content and attention/emotion adaptations with the ICT syllabus.	Categorized videos by grade and syllabus topic to support topic-relevant adaptations.	
Improve hand/finger visibility and organize ICT signs by syllabus.	Added 3D sign models for clearer visualization and organized signs by ICT syllabus topics.	
Generate syllabus-specific sign-language learning plans.	Structured plans around Grade 10/11 ICT terminology and syllabus areas.	

Accessible Emotion-Aware E-Learning Platform for Deaf Students Studying O/L Information Technology

Problem

Traditional e-learning platforms are mainly designed for hearing students and commonly depend on audio explanations, spoken instructions and text-heavy learning materials. These platforms are unable to identify when a deaf learner becomes distracted, confused, drowsy or misses an important learning concept.

As a result:

- Student attention loss may remain unnoticed.
- Missed lesson sections are not automatically identified.
- Learning support is usually static and not personalized.
- Generic quizzes are not connected to the exact concepts a student missed.
- Technical ICT concepts may be difficult to understand using text alone.
- Traditional audio alerts are unsuitable for deaf learners.
- Teachers have limited visibility into individual learning difficulties.

Adaptive Text-to-Sign Avatar and Transcript-Based Learning Support System

- Converts lesson text into sign-language animations using a 3D avatar.
- Displays clear hand and finger movements.
- Provides lesson transcripts to support reading and revision.
- Helps Deaf students learn through signs and text.



Mendis L B L | IT22197764

NLP-Based Emotion-Aware and Intent-Aware Adaptive Chatbot



Kothalawala L S S | IT22185884

- Identifies the student's learning need and relevant ICT syllabus topic.
- Uses **EARA** to adapt explanations based on feedback, understanding, and available attention/emotion indicators.
- Provides simple explanations, analogies, and exam guidance.
- Offers short notes, quizzes, and micro-challenges for revision.
- Tracks progress and highlights repeated learning difficulties for teachers.

Smart Haptic Wristband & Sign Practice Arena

- Helps students practise ICT-related signs through interactive activities.
- Uses a webcam to track hand positions and check supported signs.
- Provides visual feedback and records practice progress.
- Includes an ESP32 wristband prototype with vibration alerts and a small display.



Nawarathna I K K R B | IT19131948

Real-Time Emotion Detection and Attention Monitoring



Tharaka W A K N | IT22211200

- Uses webcam input to detect facial expressions and monitor attention.
- Automatically pauses and resumes lessons based on attention.
- Supports hand gestures to control lesson playback.
- Shares emotion indicators with the chatbot for adaptive support.
- Generates attention reports for teachers.

User Feedback:

- Explanations are easy to follow.
- Short notes help with revision
- Sign animations make lessons easier to follow.
- Automatic pausing helps avoid missing content.
- The chatbot explains difficult topics simply.
- Sign practice makes learning interactive.
- More signs and visual examples would help.

Limitations:

- Limited sign vocabulary.
- Detection depends on lighting and camera quality.
- Chatbot content focuses on Grade 10 and 11 ICT.

Future Improvements

- Expand ICT signs and lesson content.
- Improve emotion and attention detection.
- Extend chatbot support to more subjects and grades.

Thank You

Research Group: R26-IT-018